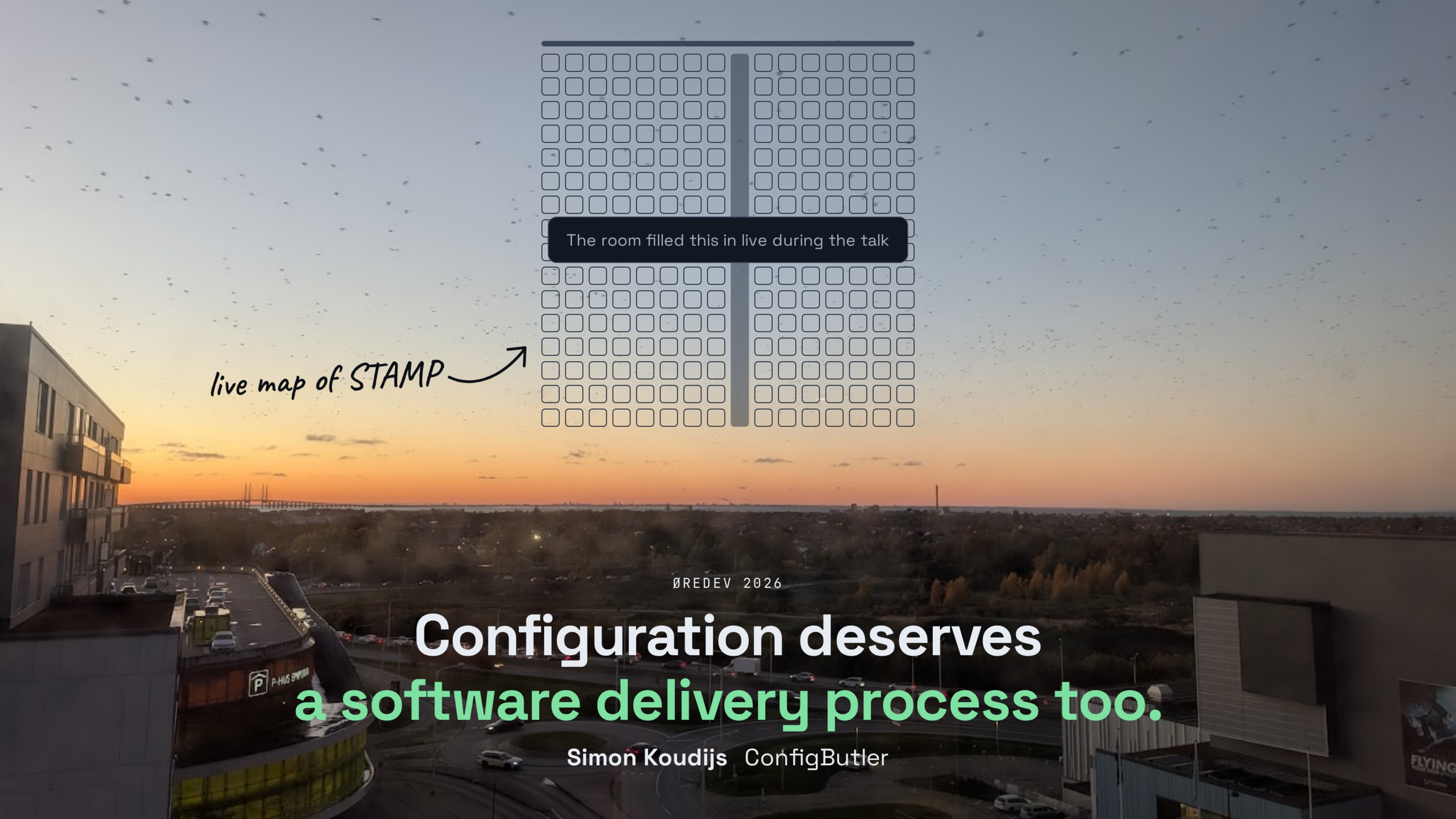


ØREDEV 2026

Configuration deserves a software delivery process too.

Simon Koudijs ConfigButler

TLDR;

slides at: **koudijs.dev/speaking**

Package settings into an image.

Settings screen → Files in Git → **Image** → Test → Production

Model config text-based

Desired state, format choice is yours: **YAML**, XML, INI

A clear place to point at

What is production running? This digest.

Promoted like code

Workloads now get 2:
executable image
config image

There is lots of room for refinement: the core works.

LINKS

- **GitOps Reverser** — github.com/ConfigButler/gitops-reverser
- **settings-image** — github.com/ConfigButler/settings-image
- **Reverse GitOps** — reversegitops.dev
- **Config Deserves Better** — configdeservesbetter.dev

What if we could make the **configuration API** so good,

and store our settings in an **audit log** so good,

that people lose the tendency to build tools in front of it?

Infra as code? Let’s go for infra as data.

TO BE CLEAR

This talk is not a bash on:

- config-as-code
- databases
- legacy applications

But I do dislike it if I just can't **hunt back what has been changing my system.**

Simon Koudijs

`[Software|Product|Cloud|AI] engineer`

- From the Netherlands
- Left my engineering job in 2025 to build my own company
- Frustrated by settings pages: do we still build that ourselves?
- **koudijs.dev**: consultancy, training, speaking
- **ConfigButler**: the hosted product of this idea

infra + config + code

= production

WHICH CONFIGURATION DO I MEAN?

Is this infra?

YOUR PLATFORM TEAM CHANGES IT

YOUR END USERS CHANGE IT

INFRA

- Servers
- Networks
- Certificates

ENVIRONMENT-SPECIFIC

- Hostnames
- Replica counts
- Log levels

differs per environment

CONNECTION STRINGS

- To your databases

how the app reaches its own data

END-USER SELF-SERVICE

- Your chair's colour
- Feature flags
- Business rules
- API keys for external services

changed live, by people using your system

TRANSACTION DATA

- Orders
- Payments

what your database is for

Do that like you are used to.

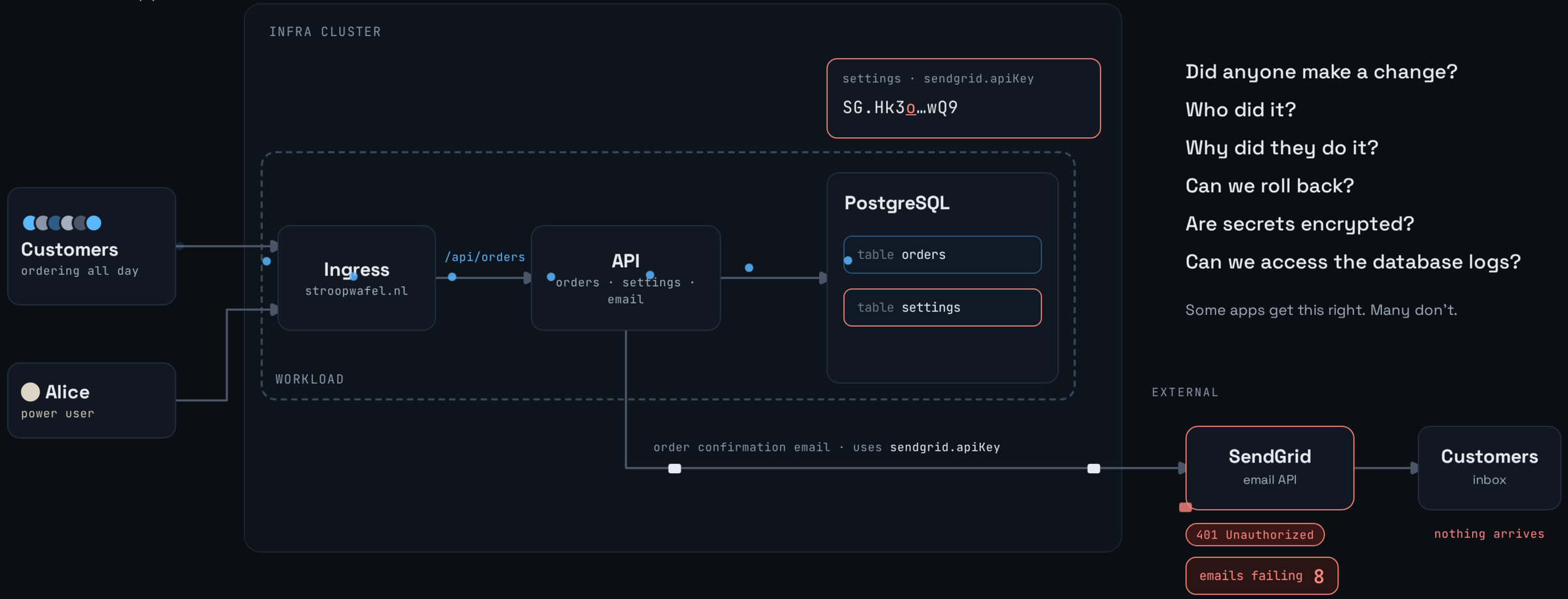
This talk.

Let's agree not to talk about that today.

THE PROBLEM

One typo. No more customer emails.

Settings often live in the app's database, next to the orders. One small edit can break production, and it is not always clear what happened.



Did anyone make a change?

Who did it?

Why did they do it?

Can we roll back?

Are secrets encrypted?

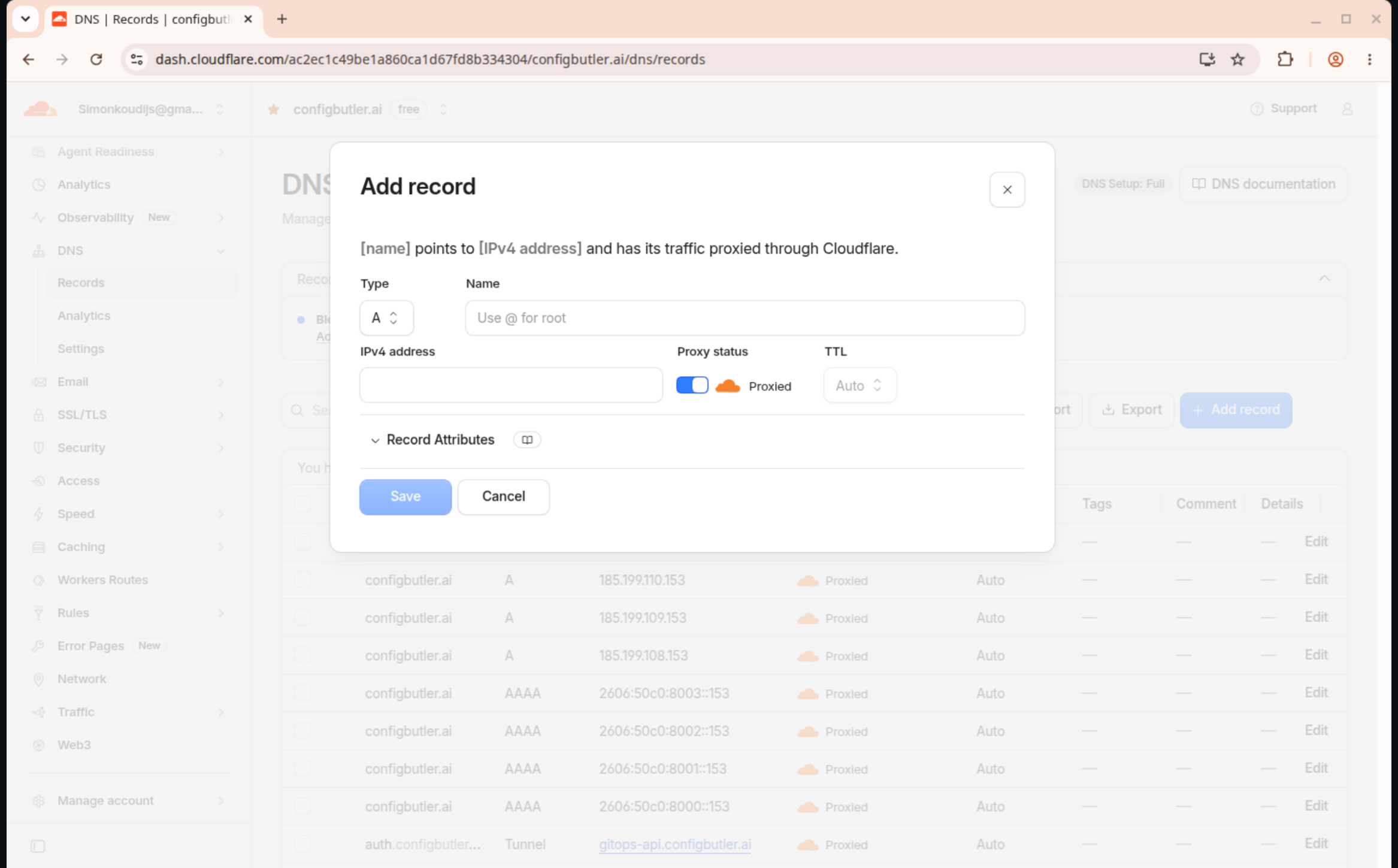
Can we access the database logs?

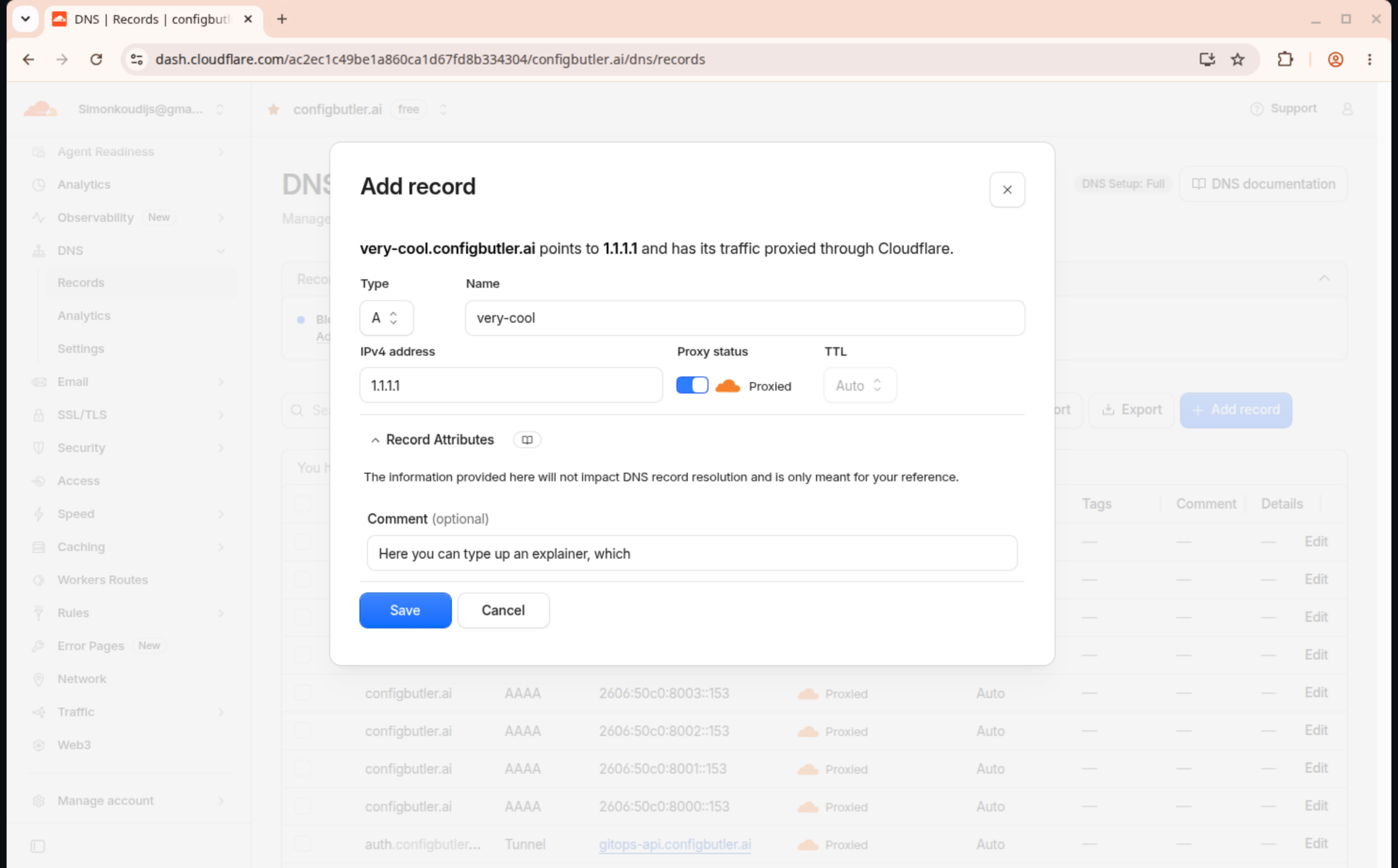
Some apps get this right. Many don't.

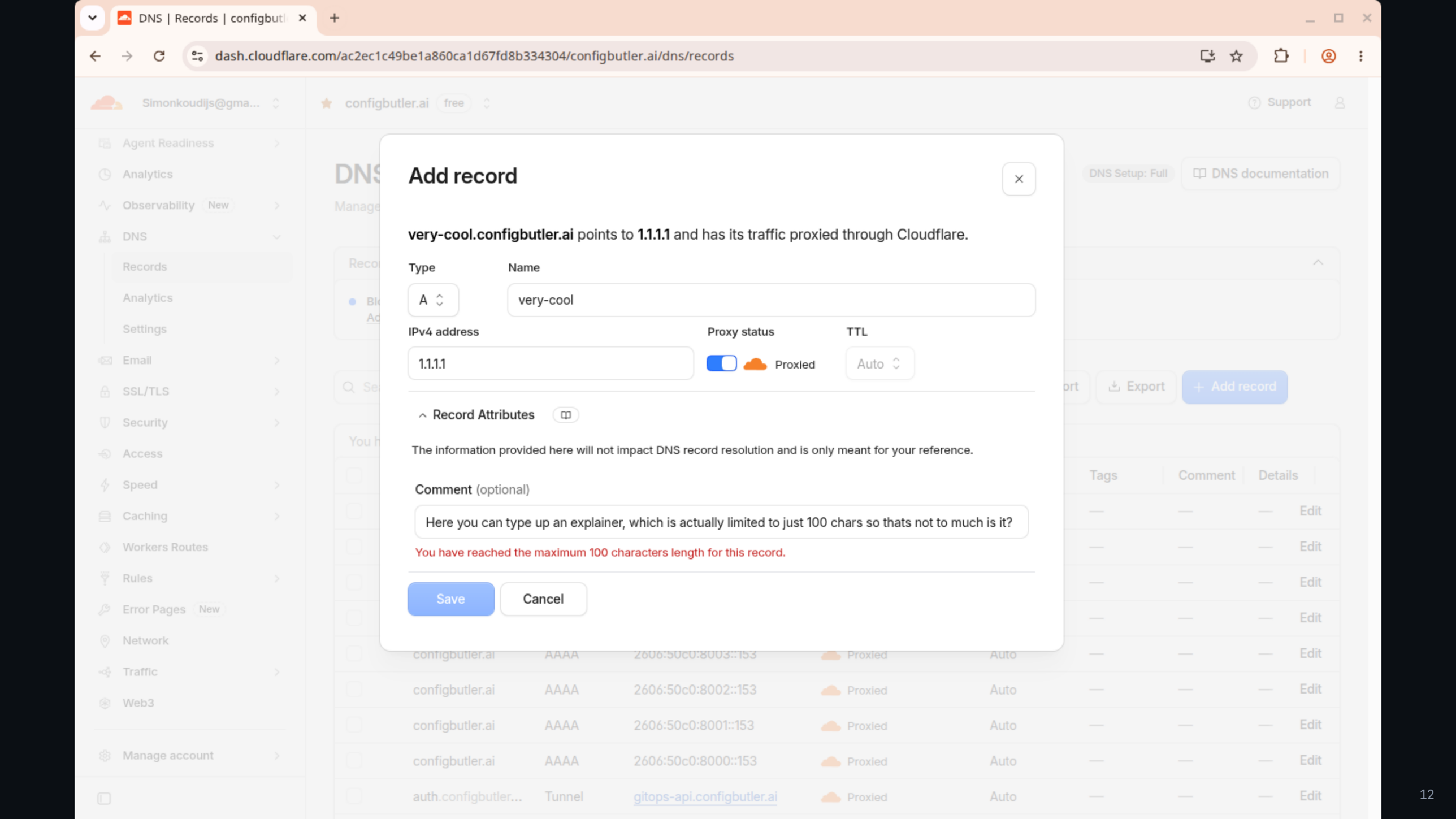
04 Some apps handle this well. Many don't, and you find out in production.

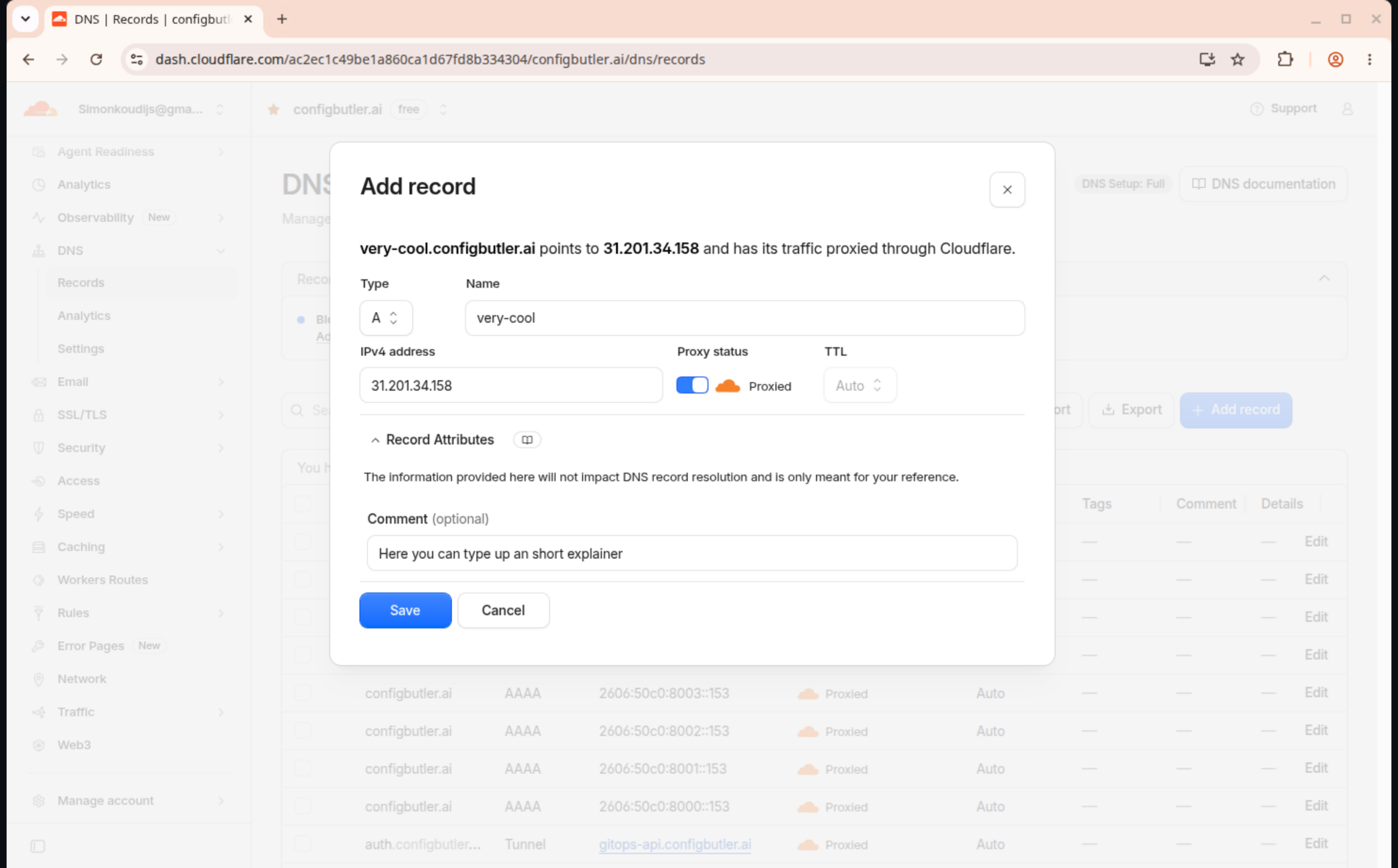
01 emails flow · 02 one typo · 03 silent failure · 04 no answers

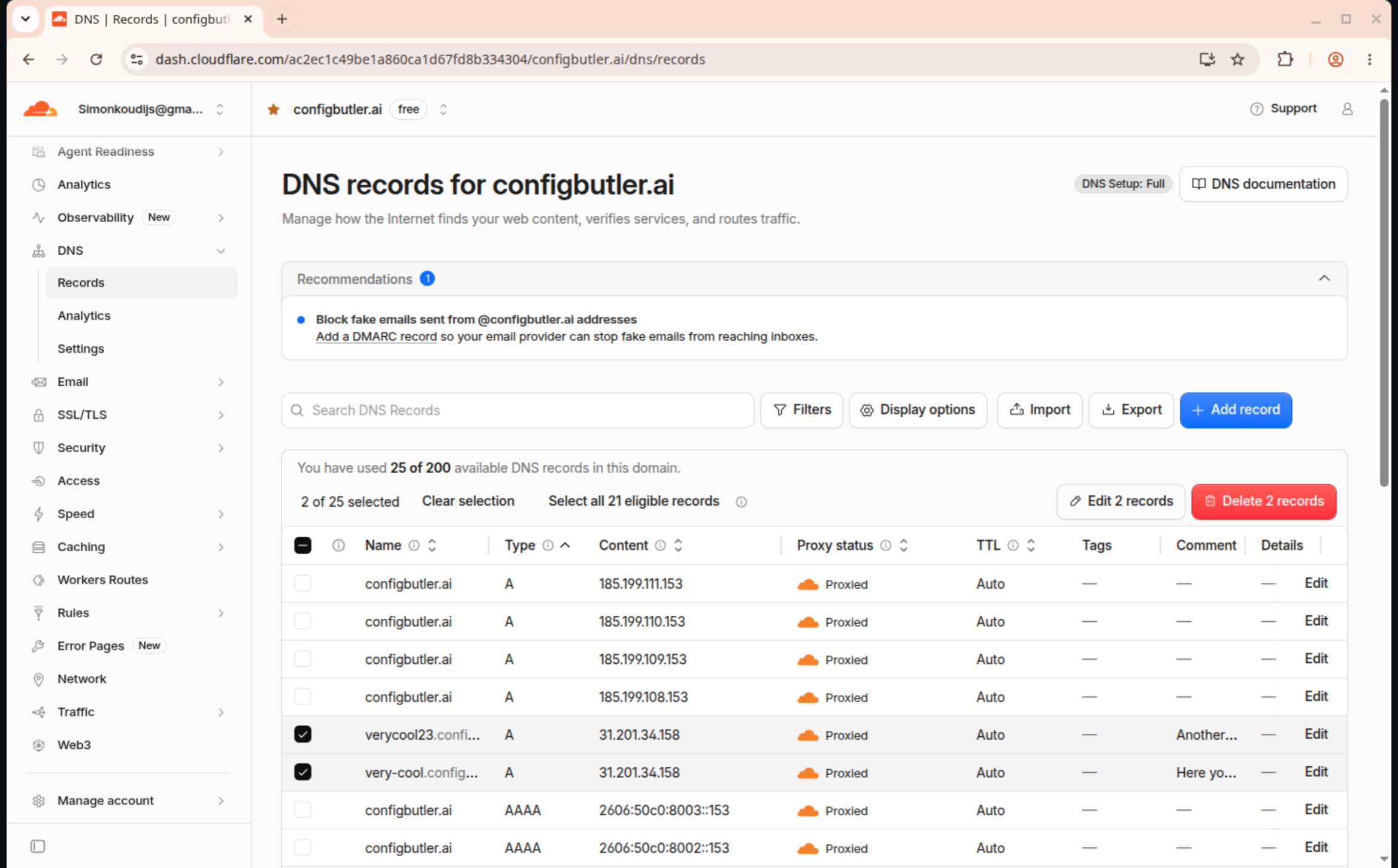
Settings screen
Files in Git
Image
Test
Production

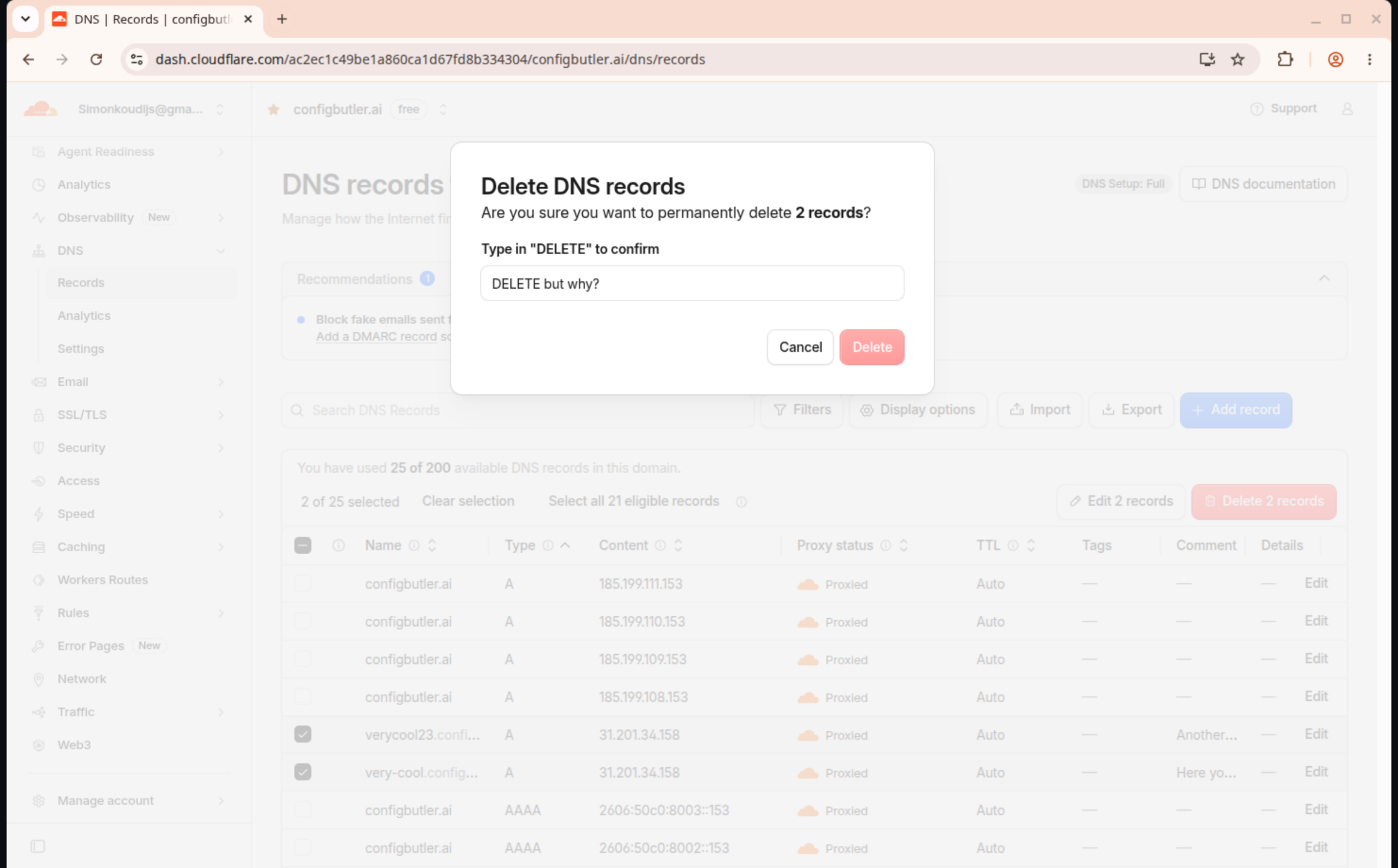

DNS | Records | configbutl
dash.cloudflare.com/ac2ec1c49be1a860ca1d67fd8b334304/configbutler.ai/dns/records
Simonkoudijs@gma...
configbutler.ai
free
Support
Agent Readiness
Analytics
Observability
New
DNS
Records
Analytics
Settings
Email
SSL/TLS
Security
Access
Speed
Caching
Workers Routes
Rules
Error Pages
New
Network
Traffic
Web3
Manage account
DNS
Manage
DNS Setup: Full
DNS documentation
Export
Add record
Tags
Comment
Details
Add record
[name] points to [IPv4 address] and has its traffic proxied through Cloudflare.
Type
A
Name
Use @ for root
IPv4 address
Proxy status
Proxied
TTL
Auto
Record Attributes
Save
Cancel
configbutler.ai A 185.199.110.153 Proxied Auto Edit
configbutler.ai A 185.199.109.153 Proxied Auto Edit
configbutler.ai A 185.199.108.153 Proxied Auto Edit
configbutler.ai AAAA 2606:50c0:8003::153 Proxied Auto Edit
configbutler.ai AAAA 2606:50c0:8002::153 Proxied Auto Edit
configbutler.ai AAAA 2606:50c0:8001::153 Proxied Auto Edit
configbutler.ai AAAA 2606:50c0:8000::153 Proxied Auto Edit
auth.configbutler... Tunnel gitops-api.configbutler.ai Proxied Auto Edit


DNS | Records | configbutl
dash.cloudflare.com/ac2ec1c49be1a860ca1d67fd8b334304/configbutler.ai/dns/records
Simonkoudijs@gma...
configbutler.ai
free
Support
Agent Readiness
Analytics
Observability
New
DNS
Records
Analytics
Settings
Email
SSL/TLS
Security
Access
Speed
Caching
Workers Routes
Rules
Error Pages
New
Network
Traffic
Web3
Manage account
DNS
Manage
DNS Setup: Full
DNS documentation
Export
Add record
Tags
Comment
Details
Edit
Add record
very-cool.configbutler.ai points to 1.1.1.1 and has its traffic proxied through Cloudflare.
Type
A
Name
very-cool
IPv4 address
1.1.1.1
Proxy status
Proxied
TTL
Auto
Record Attributes
The information provided here will not impact DNS record resolution and is only meant for your reference.
Comment (optional)
Here you can type up an explainer, which
Save
Cancel
configbutler.ai AAAA 2606:50c0:8003::153 Proxied Auto
configbutler.ai AAAA 2606:50c0:8002::153 Proxied Auto
configbutler.ai AAAA 2606:50c0:8001::153 Proxied Auto
configbutler.ai AAAA 2606:50c0:8000::153 Proxied Auto
auth.configbutler... Tunnel gitops-api.configbutler.ai Proxied Auto


DNS | Records | configbutl
dash.cloudflare.com/ac2ec1c49be1a860ca1d67fd8b334304/configbutler.ai/dns/records
Simonkoudijs@gma...
configbutler.ai
free
Support
Agent Readiness
Analytics
Observability
New
DNS
Records
Analytics
Settings
Email
SSL/TLS
Security
Access
Speed
Caching
Workers Routes
Rules
Error Pages
New
Network
Traffic
Web3
Manage account
DNS Setup: Full
DNS documentation
Export
Add record
Add record
very-cool.configbutler.ai points to 1.1.1.1 and has its traffic proxied through Cloudflare.
Type
A
Name
very-cool
IPv4 address
1.1.1.1
Proxy status
Proxied
TTL
Auto
Record Attributes
The information provided here will not impact DNS record resolution and is only meant for your reference.
Comment (optional)
Here you can type up an explainer, which is actually limited to just 100 chars so thats not to much is it?
You have reached the maximum 100 characters length for this record.
Save
Cancel
Tags
Comment
Details
configbutler.ai AAAA 2606:50c0:8003::153 Proxied Auto Edit
configbutler.ai AAAA 2606:50c0:8002::153 Proxied Auto Edit
configbutler.ai AAAA 2606:50c0:8001::153 Proxied Auto Edit
configbutler.ai AAAA 2606:50c0:8000::153 Proxied Auto Edit
auth.configbutler... Tunnel gitops-api.configbutler.ai Proxied Auto Edit


DNS | Records | configbutl
dash.cloudflare.com/ac2ec1c49be1a860ca1d67fd8b334304/configbutler.ai/dns/records
Simonkoudijs@gma...
configbutler.ai
free
Support
Agent Readiness
Analytics
Observability
New
DNS
Records
Analytics
Settings
Email
SSL/TLS
Security
Access
Speed
Caching
Workers Routes
Rules
Error Pages
New
Network
Traffic
Web3
Manage account
DNS Setup: Full
DNS documentation
Export
Add record
Tags
Comment
Details
Edit
Add record
very-cool.configbutler.ai points to 31.201.34.158 and has its traffic proxied through Cloudflare.
Type
A
Name
very-cool
IPv4 address
31.201.34.158
Proxy status
Proxied
TTL
Auto
Record Attributes
The information provided here will not impact DNS record resolution and is only meant for your reference.
Comment (optional)
Here you can type up an short explainer
Save
Cancel
configbutler.ai AAAA 2606:50c0:8003::153 Proxied Auto
configbutler.ai AAAA 2606:50c0:8002::153 Proxied Auto
configbutler.ai AAAA 2606:50c0:8001::153 Proxied Auto
configbutler.ai AAAA 2606:50c0:8000::153 Proxied Auto
auth.configbutler... Tunnel gitops-api.configbutler.ai Proxied Auto


DNS | Records | configbutl
dash.cloudflare.com/ac2ec1c49be1a860ca1d67fd8b334304/configbutler.ai/dns/records
Simonkoudijs@gma...
configbutler.ai
free
Support
Agent Readiness
Analytics
Observability
New
DNS
Records
Analytics
Settings
Email
SSL/TLS
Security
Access
Speed
Caching
Workers Routes
Rules
Error Pages
New
Network
Traffic
Web3
Manage account
DNS records for configbutler.ai
Manage how the Internet finds your web content, verifies services, and routes traffic.
DNS Setup: Full
DNS documentation
Recommendations
1
Block fake emails sent from @configbutler.ai addresses
Add a DMARC record so your email provider can stop fake emails from reaching inboxes.
Search DNS Records
Filters
Display options
Import
Export
Add record
You have used 25 of 200 available DNS records in this domain.
2 of 25 selected
Clear selection
Select all 21 eligible records
Edit 2 records
Delete 2 records
Name
Type
Content
Proxy status
TTL
Tags
Comment
Details
configbutler.ai A 185.199.111.153 Proxied Auto — — — Edit
configbutler.ai A 185.199.110.153 Proxied Auto — — — Edit
configbutler.ai A 185.199.109.153 Proxied Auto — — — Edit
configbutler.ai A 185.199.108.153 Proxied Auto — — — Edit
verycool23.confi... A 31.201.34.158 Proxied Auto — Another... — Edit
very-cool.config... A 31.201.34.158 Proxied Auto — Here yo... — Edit
configbutler.ai AAAA 2606:50c0:8003::153 Proxied Auto — — — Edit
configbutler.ai AAAA 2606:50c0:8002::153 Proxied Auto — — — Edit


DNS | Records | configbutl
dash.cloudflare.com/ac2ec1c49be1a860ca1d67fd8b334304/configbutler.ai/dns/records
Simonkoudijs@gma...
configbutler.ai
free
Support
Agent Readiness
Analytics
Observability
New
DNS
Records
Analytics
Settings
Email
SSL/TLS
Security
Access
Speed
Caching
Workers Routes
Rules
Error Pages
New
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Traffic
Web3
Manage account
DNS records
Manage how the Internet fir
DNS Setup: Full
DNS documentation
Recommendations
Block fake emails sent f
Add a DMARC record sc
Search DNS Records
Filters
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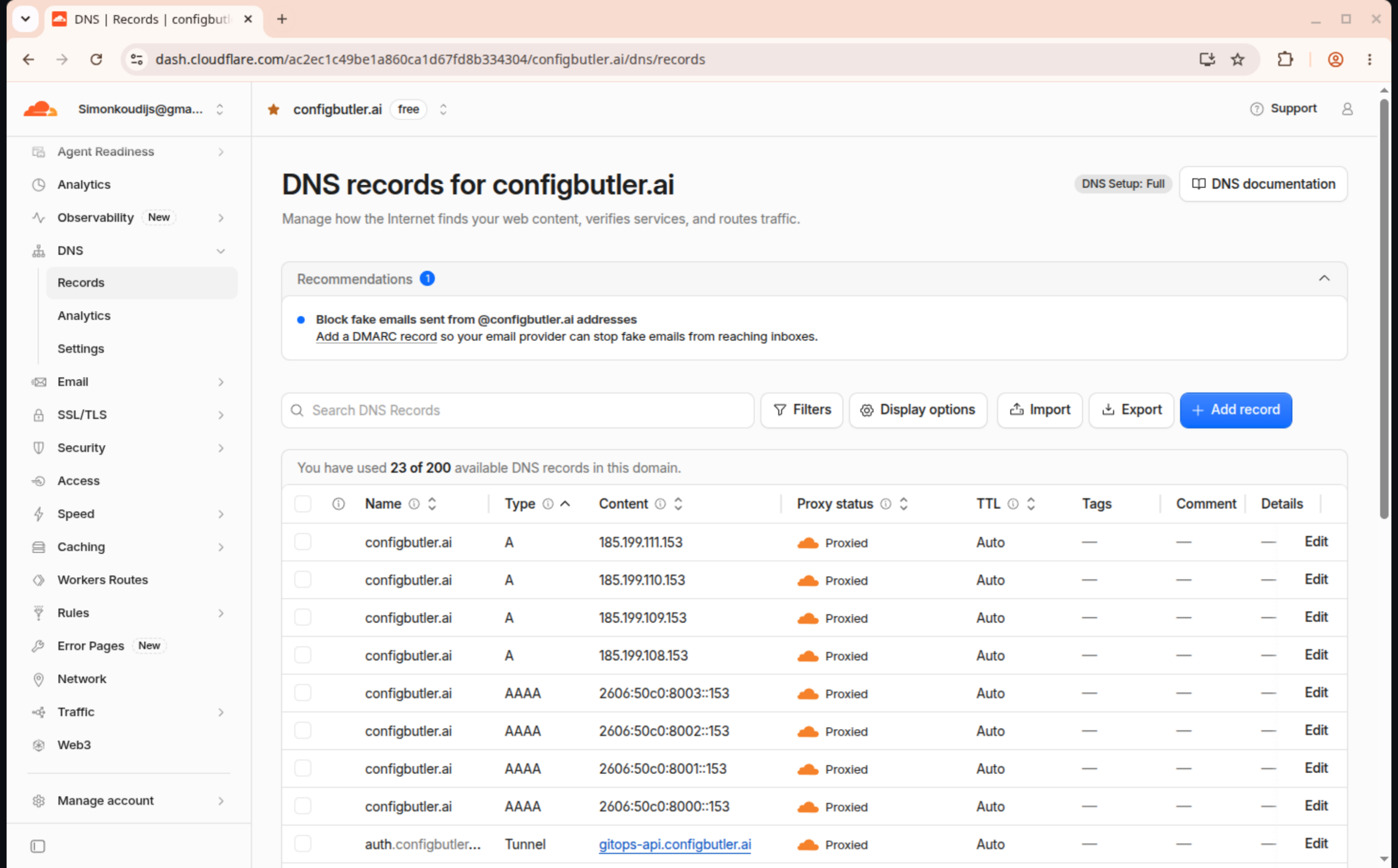
You have used 25 of 200 available DNS records in this domain.
2 of 25 selected
Clear selection
Select all 21 eligible records
Edit 2 records
Delete 2 records
Name
Type
Content
Proxy status
TTL
Tags
Comment
Details
configbutler.ai A 185.199.111.153 Proxied Auto Edit
configbutler.ai A 185.199.110.153 Proxied Auto Edit
configbutler.ai A 185.199.109.153 Proxied Auto Edit
configbutler.ai A 185.199.108.153 Proxied Auto Edit
verycool23.confi... A 31.201.34.158 Proxied Auto Another... Edit
very-cool.config... A 31.201.34.158 Proxied Auto Here yo... Edit
configbutler.ai AAAA 2606:50c0:8003::153 Proxied Auto Edit
configbutler.ai AAAA 2606:50c0:8002::153 Proxied Auto Edit
Delete DNS records
Are you sure you want to permanently delete 2 records?
Type in "DELETE" to confirm
DELETE but why?
Cancel
Delete


DNS | Records | configbutl
dash.cloudflare.com/ac2ec1c49be1a860ca1d67fd8b334304/configbutler.ai/dns/records
Simonkoudijs@gma...
configbutler.ai
free
Support
Agent Readiness
Analytics
Observability
New
DNS
Records
Analytics
Settings
Email
SSL/TLS
Security
Access
Speed
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Workers Routes
Rules
Error Pages
New
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Traffic
Web3
Manage account
DNS records for configbutler.ai
Manage how the Internet finds your web content, verifies services, and routes traffic.
DNS Setup: Full
DNS documentation
Recommendations
1
Block fake emails sent from @configbutler.ai addresses
Add a DMARC record so your email provider can stop fake emails from reaching inboxes.
Search DNS Records
Filters
Display options
Import
Export
Add record
You have used 23 of 200 available DNS records in this domain.
Name
Type
Content
Proxy status
TTL
Tags
Comment
Details
configbutler.ai A 185.199.111.153 Proxied Auto — — — Edit
configbutler.ai A 185.199.110.153 Proxied Auto — — — Edit
configbutler.ai A 185.199.109.153 Proxied Auto — — — Edit
configbutler.ai A 185.199.108.153 Proxied Auto — — — Edit
configbutler.ai AAAA 2606:50c0:8003::153 Proxied Auto — — — Edit
configbutler.ai AAAA 2606:50c0:8002::153 Proxied Auto — — — Edit
configbutler.ai AAAA 2606:50c0:8001::153 Proxied Auto — — — Edit
configbutler.ai AAAA 2606:50c0:8000::153 Proxied Auto — — — Edit
auth.configbutler... Tunnel gitops-api.configbutler.ai Proxied Auto — — — Edit

Audit logs | Simonkoudijs@

dash.cloudflare.com/ac2ec1c49be1a860ca1d67fd8b334304/audit-log?action_type~neq=view

Simonkoudijs@gma...

Audit logs

Support

configbutler.ai / DNS

configbutler.ai
Domains

Audit logs
Manage account

Analytics
koudijs.dev / DNS

Observability New

Email

SSL/TLS

Web3

Manage account

Members

Billing

Account API tokens

Audit logs

Alerts

Blocked content

Abuse reports

Carbon Impact Report

Configurations

Tagged Resources Beta

Actio... ≠ view × + Add filter Clear filters

Last 7 days

Actions Over Time

25 Success 0 Failed

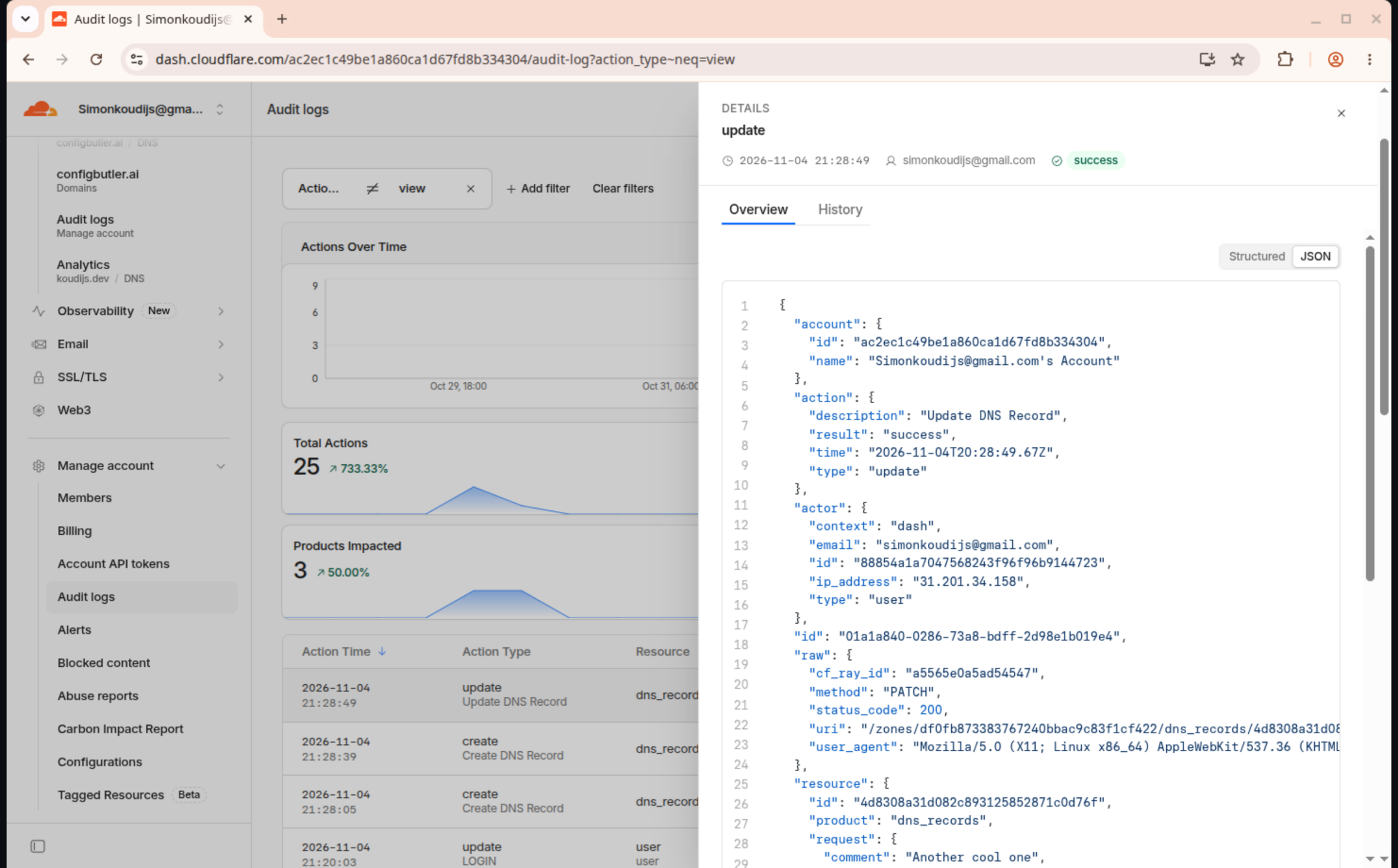
Total Actions
25 ↗ 733.33%

Unique Actors
3 ↗ 200.00%

Products Impacted
3 ↗ 50.00%

Failure Rate
0 %

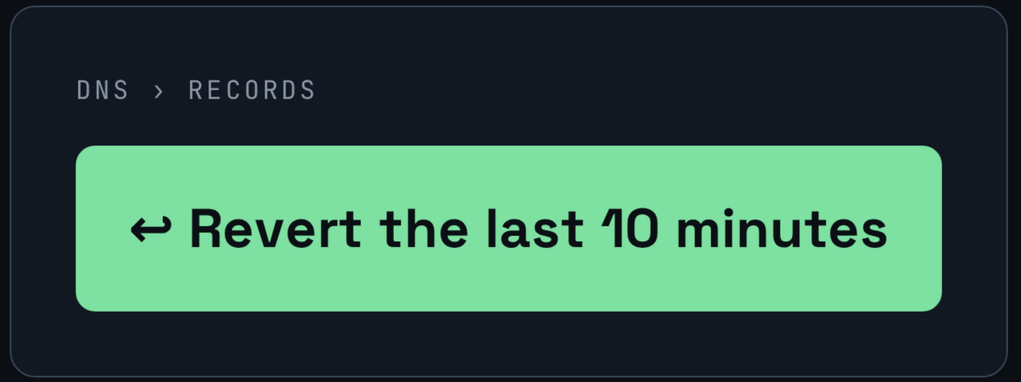
Action Time	Action Type	Resource	Actor Email	Actor Context	Zone Name
2026-11-04 21:28:49	update Update DNS Record	dns_records	simonkoudijs@gmail.com User	Dashboard	configbutler.ai
2026-11-04 21:28:39	create Create DNS Record	dns_records	simonkoudijs@gmail.com User	Dashboard	configbutler.ai
2026-11-04 21:28:05	create Create DNS Record	dns_records	simonkoudijs@gmail.com User	Dashboard	configbutler.ai
2026-11-04 21:20:03	update LOGIN	user user	simonkoudijs@gmail.com User	API (no credential detail)	-

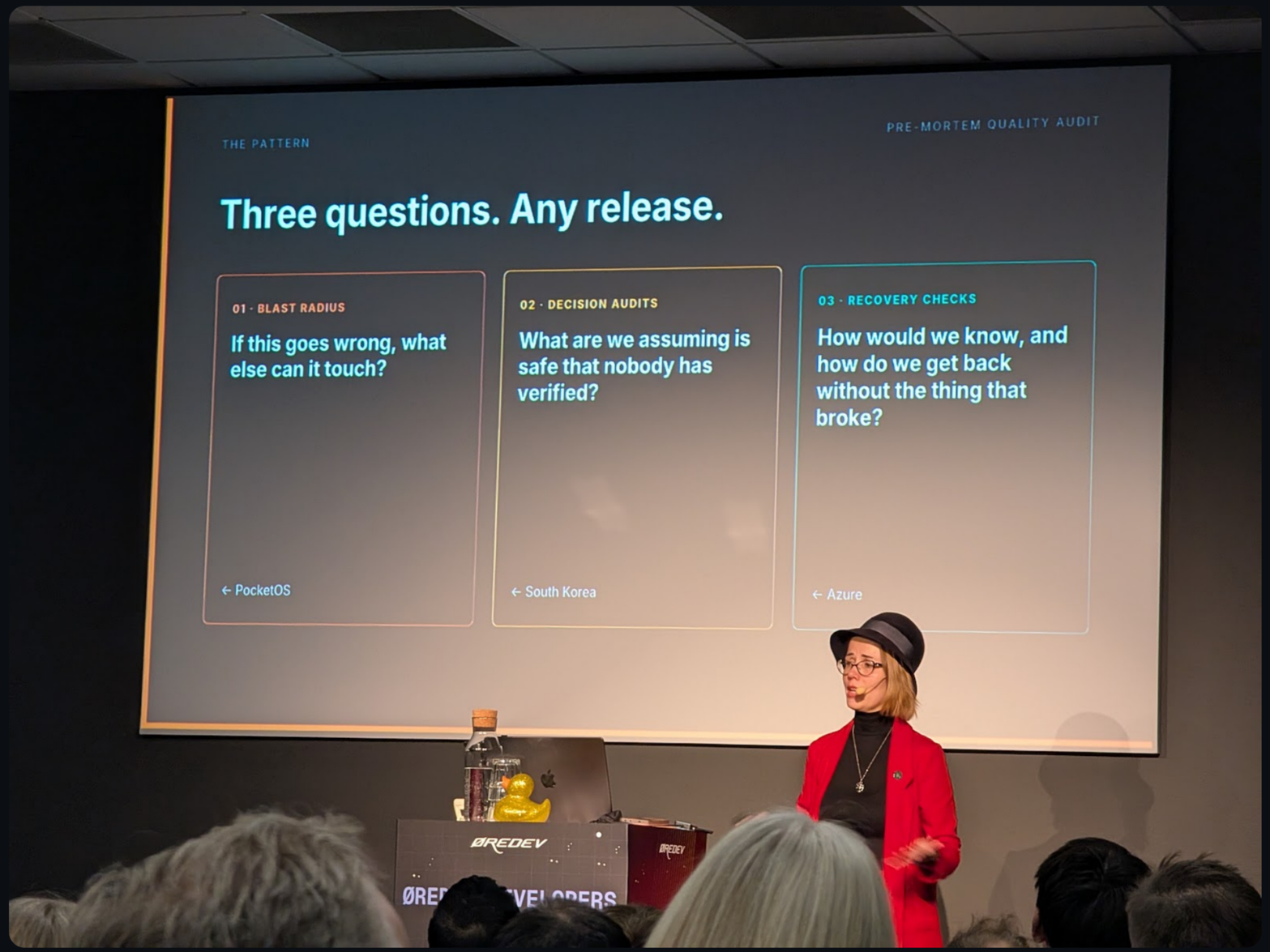

Audit logs | Simonkoudijs
dash.cloudflare.com/ac2ec1c49be1a860ca1d67fd8b334304/audit-log?action_type~neq=view
Simonkoudijs@gma...
Audit logs
configbutler.ai
Domains
Audit logs
Manage account
Analytics
koudijs.dev / DNS
Observability New
Email
SSL/TLS
Web3
Manage account
Members
Billing
Account API tokens
Audit logs
Alerts
Blocked content
Abuse reports
Carbon Impact Report
Configurations
Tagged Resources Beta
Actio... ≠ view
+ Add filter
Clear filters
Actions Over Time
9
6
3
0
Oct 29, 18:00
Total Actions
25 ↗ 733.33%
Products Impacted
3 ↗ 50.00%
Action Time
Action Type
Resource
2026-11-04 21:28:49
update
Update DNS Record
dns_record
2026-11-04 21:28:39
create
Create DNS Record
dns_record
2026-11-04 21:28:05
create
Create DNS Record
dns_record
2026-11-04 21:20:03
update
LOGIN
user
user
DETAILS
update
2026-11-04 21:28:49
simonkoudijs@gmail.com
success
Overview
History
Structured
JSON
{
"account": {
"id": "ac2ec1c49be1a860ca1d67fd8b334304",
"name": "Simonkoudijs@gmail.com's Account"
},
"action": {
"description": "Update DNS Record",
"result": "success",
"time": "2026-11-04T20:28:49.67Z",
"type": "update"
},
"actor": {
"context": "dash",
"email": "simonkoudijs@gmail.com",
"id": "88854a1a7047568243f96f96b9144723",
"ip_address": "31.201.34.158",
"type": "user"
},
"id": "01a1a840-0286-73a8-bdff-2d98e1b019e4",
"raw": {
"cf_ray_id": "a5565e0a5ad54547",
"method": "PATCH",
"status_code": 200,
"uri": "/zones/df0fb873383767240bbac9c83f1cf422/dns_records/4d8308a31d08
"user_agent": "Mozilla/5.0 (X11; Linux x86_64) AppleWebKit/537.36 (KHTML
},
"resource": {
"id": "4d8308a31d082c893125852871c0d76f",
"product": "dns_records",
"request": {
"comment": "Another cool one",

IT HAPPENS TO ME TOO

My own Cloudflare DNS change

- I could have written a better runbook
- I could have scripted the right values
- **But when I'm panicking, why isn't there a button for this?**





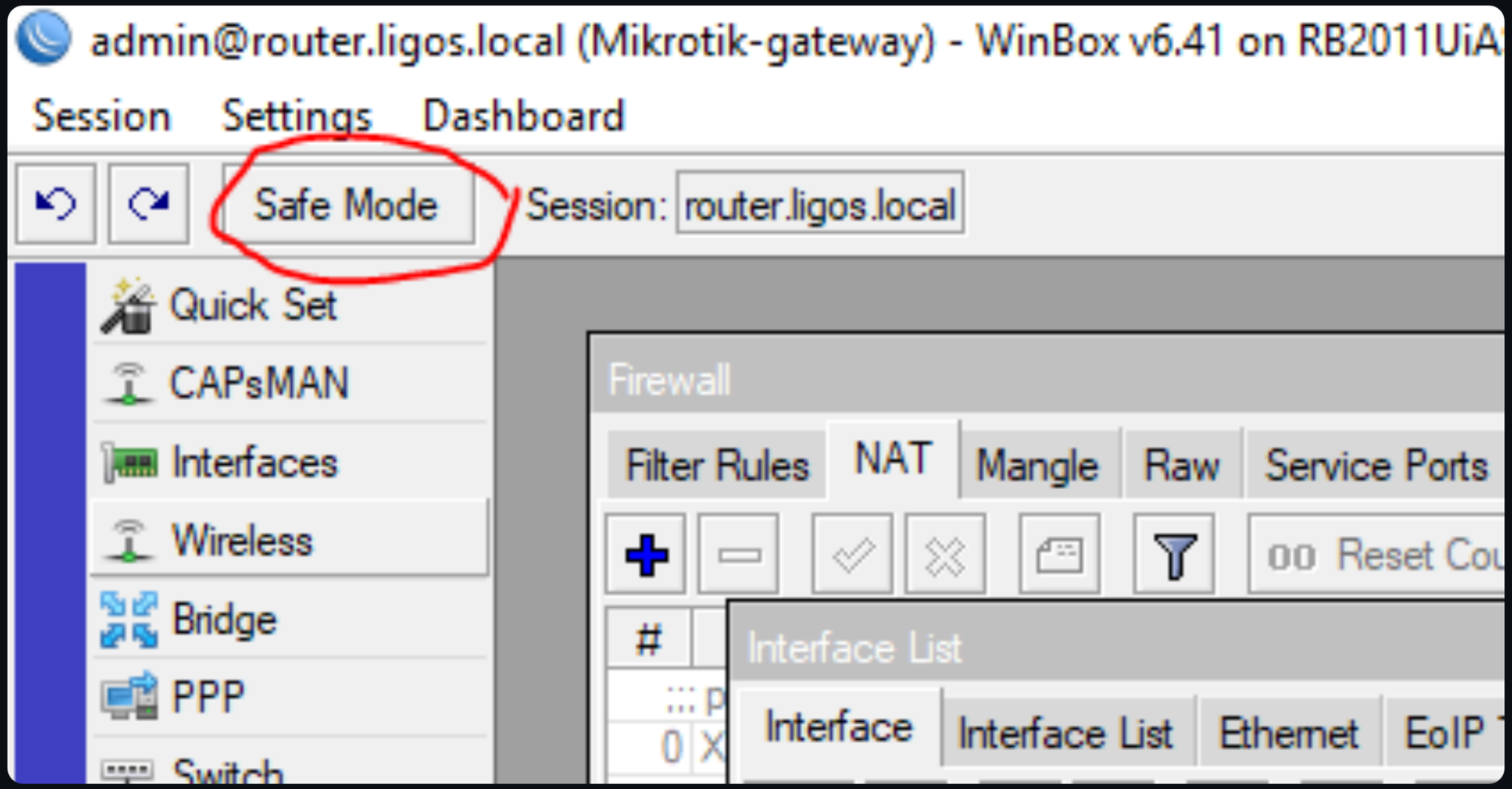
YESTERDAY, AT ØREDEV

Christine Pinto

A talk on outages.

> **“We don’t need to test that.”**

Legacy



SOURCE

Making Mikrotik Safe
blog.ligos.net/2018-02-22/Making-Mikrotik-Safe.html

Where this applies

This does NOT apply to:

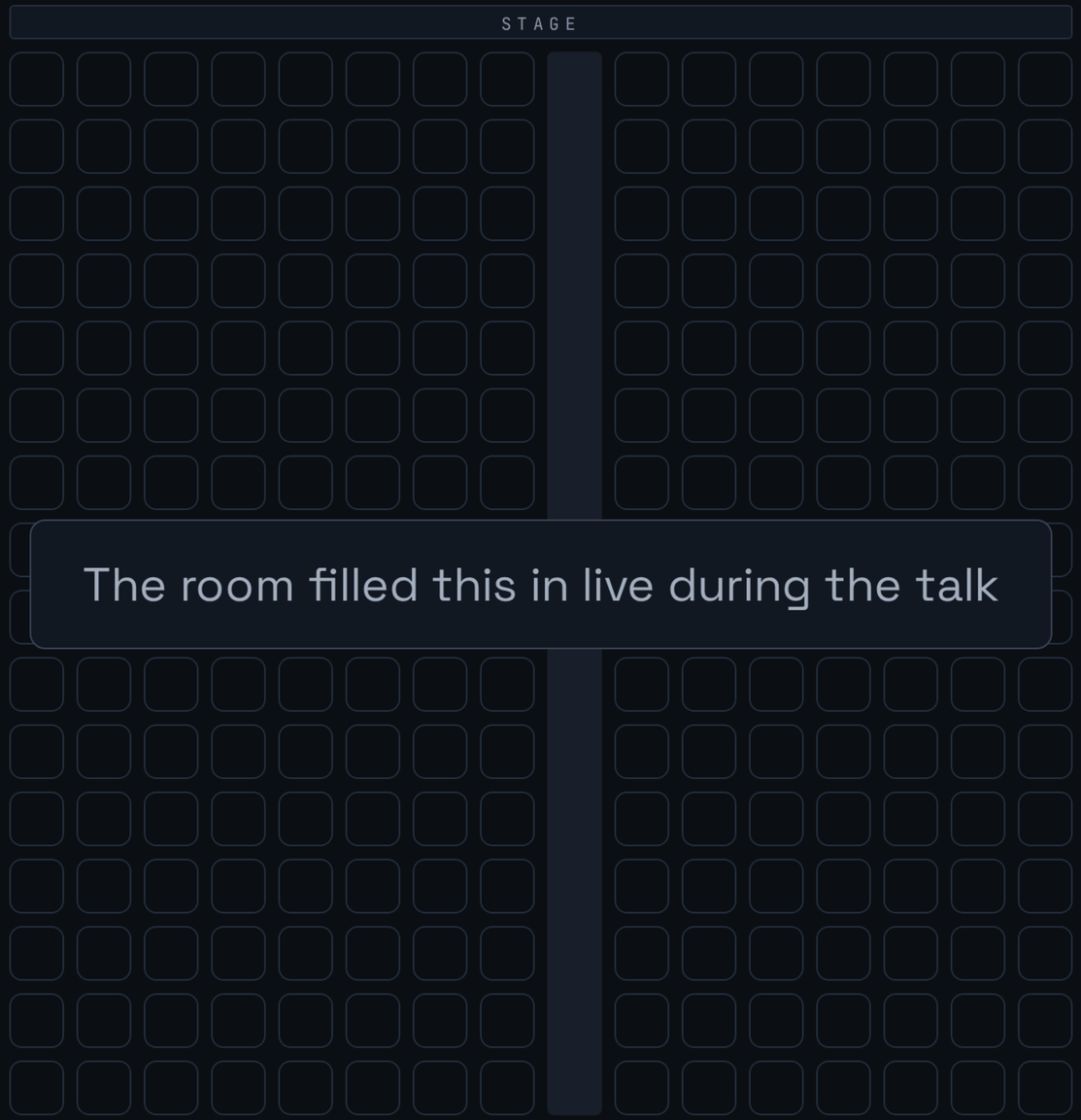
- Transactional data like new orders

This does apply to:

- Your legacy business app with a very old screen: it probably can read a file, can't it?
- Any software that needs settings and that would rather just define the things it needs

Think of it as DI for config on steroids.

Settings screen
Files in Git
Image
Test
Production


STAGE
The room filled this in live during the talk

WHAT YOU JUST DID · IN GIT

Your commits. Your names, your reasons.

```
$ cd /workspaces/talks/external/k8s-trail
$ git pull --ff-only
$ git --no-pager log --invert-grep --grep='^chore' --format='%C(cyan)%<(16,trunc)%an%C(reset) %s' -15
$ git --no-pager log -1 --invert-grep --grep='^chore' --format=fuller
```

WHAT IS IN THOSE COMMITS?

Back to text files. Oef, two steps back?

A specification

What it should be

```
row: 6
column: 6
claimedBy: simon
color: "#a8e05f"
```

- **Only the desired state.** Nothing that happened at runtime
- **Several small files that reference each other,** for what's complex
- **A special place for secret values.** We don't want them in sight

ONE CHAIR, IN ANY FORMAT

It’s all text-based files.
What format do you like?

seat.ini

```
[seat]
row = 6
column = 6
color = "#a8e05f"
```

seat.xml

```
<seat row="6" column="6">
  <color>#a8e05f</color>
</seat>
```

seat.json

```
{
  "row": 6,
  "column": 6,
  "color": "#a8e05f"
}
```

seat.yaml

```
row: 6
column: 6
color: "#a8e05f"
```

simon.jpg

even this

My suggestion: use YAML.

Oh dear, what is he proposing?

But what about my nice relational database?

And how do you handle merge conflicts?

How do we let people change that?

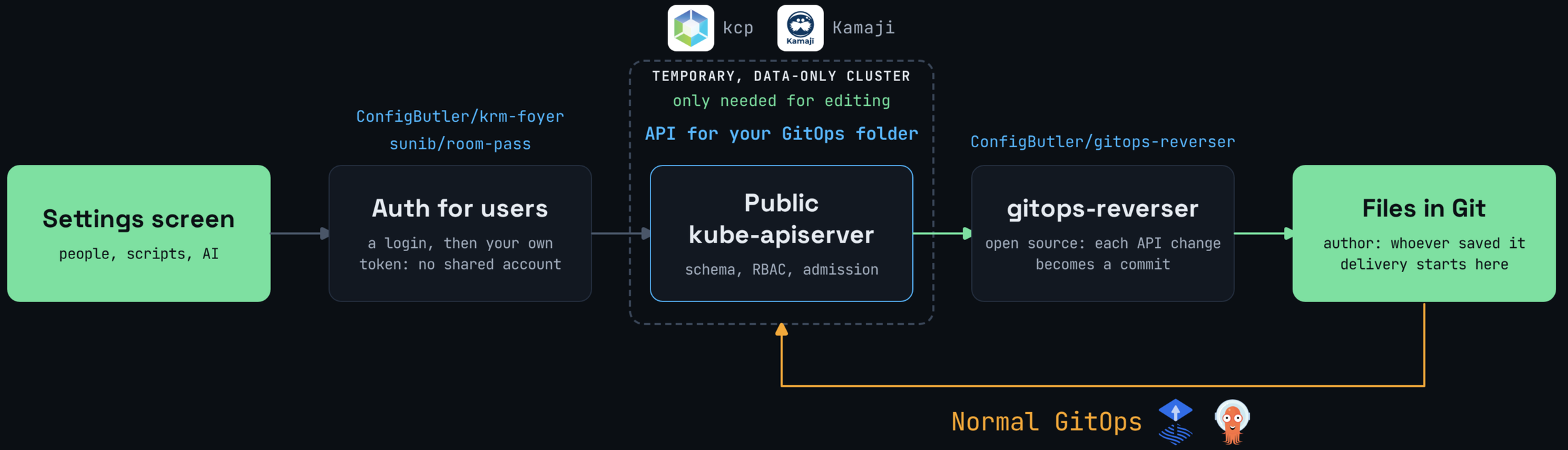
We split edits from using.

Have a very nice API for editing the files.

Hint: don't try to build that API directly on top of Git. I failed trying.

Tomorrow's TLDR;

An API and files. At the same time.



KRM makes it possible: one shape, in the API and in the files

Today I won't go deeper. Let's use it, and be happy with the magic.

04 One shape, KRM, makes this possible. Today we just use it.

FOR THE STRUCTURED ONES

One shape: KRM

```yaml
apiVersion: examples.configbutler.ai/v1alpha1
kind: Seat
metadata:
  name: ordev-1-stamp-6-6
spec:
  row: 6
  column: 6
  color: '#a8e05f'
  emoji: 🐧
```

apiVersion
which schema it follows

kind
what it is

metadata
what it's called

spec
the desired state

The Kubernetes Resource Model: a description, in a shape many tools already read.

SOPS getsops.io | age age-encryption.org

ONE OF THE POSSIBLE WAYS

A secret, encrypted in place.

coffee-mail.sops.yaml

before: we never commit this

```
kind: Secret
metadata:
  name: coffee-mail
stringData:
  apiKey: my-plain-password
```

→

coffee-mail.sops.yaml

after: the same file, only the value is encrypted

```
kind: Secret
metadata:
  name: coffee-mail
stringData:
  apiKey: ENC[AES256_GCM,data:Qdw1fCKT…
sops:
  age:
    - recipient: age1cv6yz3f0…
```

```
$ sops encrypt --in-place coffee-mail.sops.yaml
```

In place

Git still shows the shape. Only the value is hidden.

With a key

Encrypt with the public key. Only the private key decrypts.

One possible way

This is SOPS with age. It’s not the only option.

THE REAL SETTINGS OF THE COFFEE SHOP

Three files. Safe to commit.

coffeeconfig.yaml

refers to the secrets by name

```yaml
kind: CoffeeConfig
metadata:
  name: demo-coffee
spec:
  shopName: Coffee order system
  mail:
    apiKeySecretRef:
      name: coffee-mail
  payments:
    apiKeySecretRef:
      name: coffee-payments
```

→

coffee-mail.sops.yaml

```yaml
kind: Secret
name: coffee-mail
apiKey: ENC[AES256_GCM,data:Qdw1fCKT…
```

→

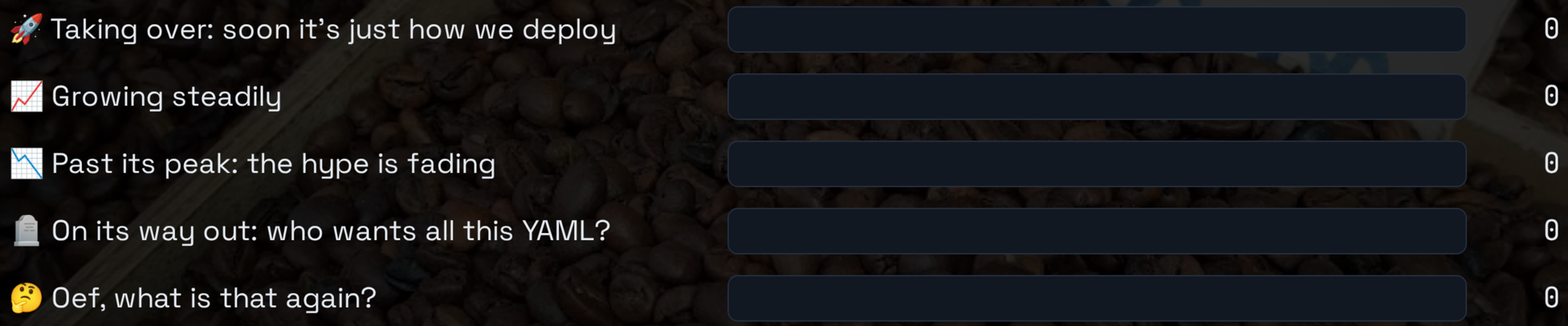
coffee-payments.sops.yaml

```yaml
kind: Secret
name: coffee-payments
apiKey: ENC[AES256_GCM,data:m93YmS91…
```

The config says which secret it needs. The secret itself stays encrypted.

WHAT YOU ANSWERED AT THE DOOR

Where is GitOps going?



The room answered this live during the talk

> **End users don’t need to be in Git.**
> **But their breaking actions should be.**

ConfigButler skills `github.com/ConfigButler/skills` **Reverse GitOps** `reversegitops.dev`

SOME OF YOU RECOGNISED THIS

Don’t dismiss this if you dislike Kubernetes.

Strong modelling

Typed objects, validation rules, and references between them.

Authz / RBAC

Already built in: who may change what, per kind and per namespace.

A well-known format

Many tools already read it, so there’s a lot available to help you.

AI can help

It knows this format. My skill reviews your model:

`github.com/ConfigButler/skills`

My dream: so common that you can buy these steps as a managed service (or self-host), while your data stays in your own Git repo.

Tomorrow, 11:15, Turing

(Ab)using the Kubernetes API as a Configuration Backend

Teaser: **bi-directional GitOps**

Settings screen
Files in Git
Image
Test
Production

settings-image github.com/ConfigButler/settings-image

coffee-settings github.com/ConfigButler/coffee-settings

FROM GIT TO AN IMAGE

From a commit to an image: just a CI step.

```yaml
on:
  push:
    branches: [main]
    paths: [coffee/**]

jobs:
  publish:
    steps:
      - uses: actions/checkout@v7
        with:
          fetch-depth: 0
      - uses: ConfigButler/settings-image@v0.2.2
        with:
          path: coffee
          repository: cr.k8s.koudijs.dev/configbutler/coffee-settings
```

A COMMIT

7fb011e

a change in the coffee/ folder

↓

CI

build and push

the same kind of job you run for code

↓

AN IMAGE

coffee-settings:main-1793863598-7fb011e

sha256:00b28189…

Creating the artifacts

A Dockerfile

```
FROM scratch
COPY settings/ /
```

You already know it. No base image and no shell.

ORAS

```
oras push \
  registry/shop-settings:43 \
  settings.yaml
```

Pushes plain files to any OCI registry.

settings-image

The same files give the same digest. Nothing in it executes. A Secret must be SOPS-encrypted, or the build stops.

Mine, and still v0.1.

OPEN SOURCE

settings-image
github.com/ConfigButler/settings-image

ORAS
oras.land

coffee-settings github.com/ConfigButler/coffee-settings

settings-image github.com/ConfigButler/settings-image

THE MAIN POINT

Your files. One image.

coffeeconfig.yaml

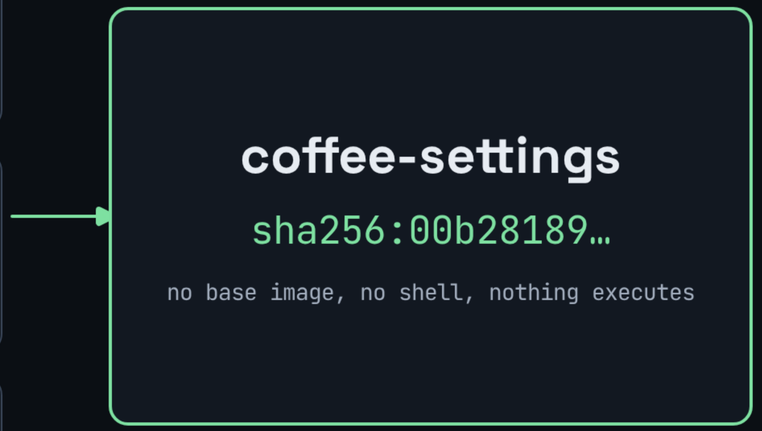
```
kind: CoffeeConfig
shopName: Coffee order system
mail:
  apiKeySecretRef:
    name: coffee-mail
```

coffee-mail.sops.yaml

```
kind: Secret
name: coffee-mail
apiKey: ENC[AES256_GCM,data:Qdw1fCKT…
```

coffee-payments.sops.yaml

```
kind: Secret
name: coffee-payments
apiKey: ENC[AES256_GCM,data:m93YmS91…
```



The complete model

of the current desired state, as your end users configured it

A clear place to point at

“What is production running?”

This digest.

04 One line answers what’s running. It also answers what was running an hour ago.

DEMO

One secret word.
Let’s follow it to the pod.

Settings screen
Files in Git
Image
Test
Production

settings-image github.com/ConfigButler/settings-image · **Kargo** kargo.io · **zot** zotregistry.dev

WHAT IS HAPPENING RIGHT NOW

Ship settings like images.

Settings ship as an image. Kargo moves it one stage at a time, and the Deployment mounts it as a read-only image volume.

Inside main-1793873184-7a53b5b · mounted at /settings

```
/settings/
└── resources/
    ├── coffeeconfig.yaml
    ├── coffee-mail.sops.yaml              # still SOPS-encrypted
    └── coffee-payments.sops.yaml          # still SOPS-encrypted

# resources/coffeeconfig.yaml
spec: { shopName: Coffee order system, mail: { apiKeySecretRef: { name: coffee-mail } } }
```

The app: Voter · just reads the files in that folder

```
# demo-kargo/app/app.yaml: how Voter reads it
SETTINGS_DIR: /settings/resources       # reads coffeeconfig.yaml
SETTINGS_KEYS: openbao-age              # age key from OpenBao opens both in memory
automountServiceAccountToken: false     # no Kubernetes API at all

# no API calls for settings: just files
```

ConfigButler/k8s · the real truth

`kargo test mounts main-1793873184-7a53b5b`

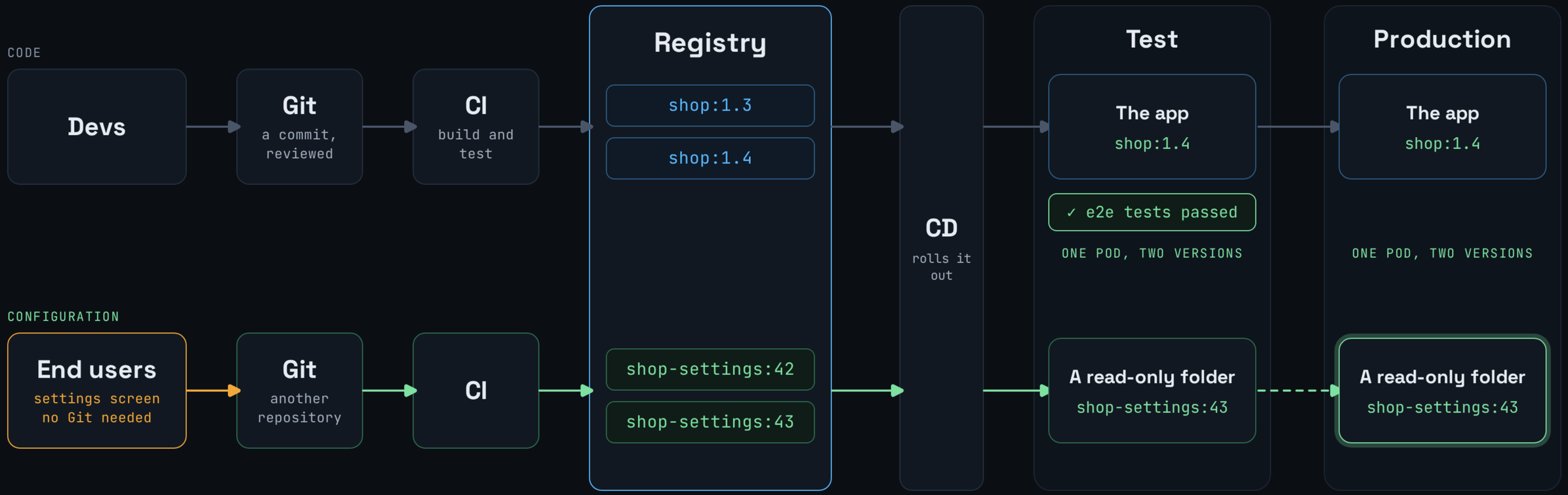
```
# demo-kargo/app/app.yaml
containers:
  - name: voter
    volumeMounts:
      - name: settings
        mountPath: /settings

# demo-kargo/test/settings.yaml: Kargo writes this, by itself
volumes:
  - name: settings
    image:
      reference: cr.k8s.koudijs.dev/configbutler/coffee-settings:main-1793873184-7a53b5b@sha256:00b28189…

# demo-kargo/production/settings.yaml: only moves when you click Promote
volumes:
  - name: settings
    image:
      reference: cr.k8s.koudijs.dev/configbutler/coffee-settings:main-1793872606-1eed2d5@sha256:fc3e12dc…
```

Cluster Flux follows that repo — `new test pod mounts main-1793873184-7a53b5b`

05 And Voter just reads those files, and opens the two secrets in memory.

01 publish · 02 registry · 03 Kargo writes · 04 mounted · 05 the app reads it

A second stream next to the code



08 No new build of the code. The pod states exactly what it runs: shop:1.4 and shop-settings:43.

DEMO TIME

This runs live. Look it up yourself.

1. A change is saved in the room
2. It becomes a commit, then an image
3. Test takes it by itself
4. Production waits until a person clicks Promote

Kargo · the tab I'll show you

`kargo.k8s.koudijs.dev/project/demo-kargo`

The room

`demo.koudijs.dev/room`

make the change here

The commits

`github.com/ConfigButler/coffee-settings`

its workflow builds the image

Test

`demo-test.koudijs.dev`

image on /dont-do-this

Production

`demo-production.koudijs.dev`

image on /dont-do-this

NONE OF THESE PARTS ARE EASY

What I haven’t solved

1. **Different values per environment**
   One image per overlay, and you no longer promote the same thing. Values from outside, and the model isn’t complete
2. **Testing it as a whole**
   One file can be checked. Does it work with this code and this infra? That takes a real environment
3. **Code and config moving apart**
   Two streams, two versions. Which combinations are valid?
4. **Going back while the API disagrees**
   Roll back the image, and the settings screen still holds the new value. Which one is the truth?
5. **Secrets over time**
   Rolling back to a key you already revoked doesn’t help

Hard questions. But with a settings table you often can’t even ask them: there is no version to point at.

Not for everything.

Leave it

- Orders and other transactions: that's what your database is for
- Settings that change every minute

Still a pain

- You need to update both your frontend and backend
- Upgrading your model: very good support, but still a pain

Worth it when

- One bad change can take production down
- You can't say who changed what, or when
- You can't say how to go back
- You're required to have much more auditing
- Settings per tenant, at scale

Coming up: EU Cyber Resilience Act

Each security update stays available for at least 10 years (Art. 13(9)). Main duties from 11 Dec 2027. So expect to keep images around for much longer.

TLDR;

slides at: **koudijs.dev/speaking**

Package settings into an image.

Settings screen → Files in Git → **Image** → Test → Production

Model config text-based

Desired state, format choice is yours: **YAML**, XML, INI

A clear place to point at

What is production running? This digest.

Promoted like code

Workloads now get 2:
executable image
config image

Tomorrow, 11:15, Turing: (Ab)using the Kubernetes API as a Configuration Backend. Is this contrarian? And is there a chance it's right?

LINKS

GitOps Reverser
github.com/ConfigButler/gitops-reverser

settings-image
github.com/ConfigButler/settings-image

Reverse GitOps
reversegitops.dev

Config Deserves Better
configdeservesbetter.dev

ONE ASK

What’s the ingredient I’m missing?

If this appeals somewhere: connect or let me know!

Tell me **one more ingredient**. Critical questions are the most welcome. I won’t spam you. I’d just love to hear from you.

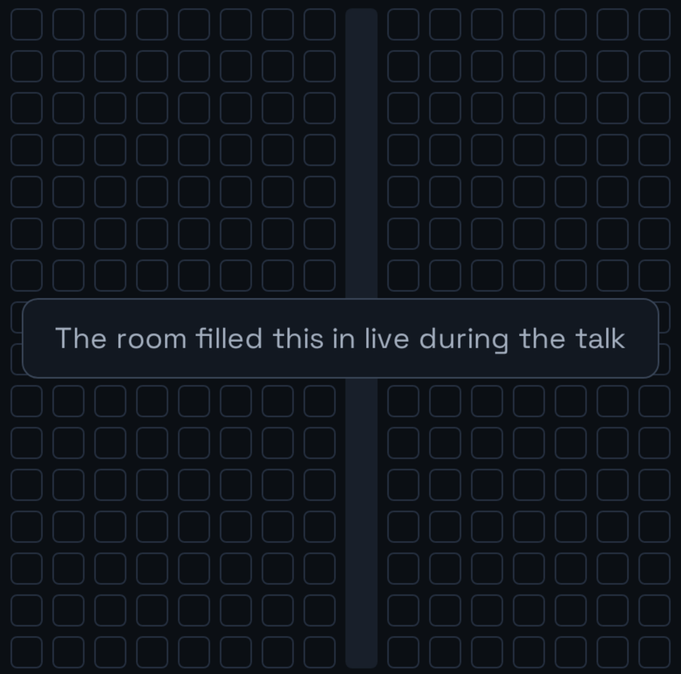
simon@koudijs.dev

The most valuable thing for me: people who give feedback.

Tomorrow, 11:15, Turing

(Ab)using the Kubernetes API as a Configuration Backend

THE ROOM · STAMP



CONNECT · LINKEDIN

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